

Maddan M

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ABOUT

Full-Stack Developer and AI/ML enthusiast with experience in computer vision, NLP, and generative AI. Passionate about building scalable web applications and deploying deep learning models. Currently exploring autonomous systems and EEG-based medical AI.

EDUCATION

Amrita School of Engineering

B.Tech in Computer and Communication Engineering – CGPA: 9.08/10.0

Chennai, India

Aug 2023 – May 2027

Jawahar CBSE School

Higher Secondary Education in Computer Science – Score: 87.5%

Neyveli, India

Jun 2021 – May 2023

EXPERIENCE

Neyveli Lignite Corporation (NLC) | Web Development Intern | HTML, CSS, JavaScript

Nov 2024 | 21 Days Onsite

- Built a **real-time dashboard** to monitor and track mining bench data, improving operational visibility for engineers.
- Designed interactive UI components for **data visualization** and reporting using front-end technologies.

PROJECTS

Unified Lightweight ADAS Architecture | YOLOv8, Deep Learning, Computer Vision, Autonomous Systems | [GitHub](#)

- Designed a real-time **ADAS** system with **Traffic Sign Recognition (TSR)** and **Forward Collision Warning (FCW)** using lightweight YOLOv8n with bimodal fusion and knowledge distillation.
- Achieved **mAP@0.5 $\approx$ 0.95** for traffic signs and **>0.81** for vehicle detection; implemented **Time-to-Collision (TTC)** estimation for hazard assessment.

Automated ADHD Detection via Cross-Channel Attention Transformers | EEG, Transformers, Medical AI | [GitHub](#)

- Proposed a novel **Cross-Channel Attention Transformer Network (CCATN)** modelling spatial dependencies across **19 EEG channels**; outperformed ViT baseline with fewer parameters.
- Achieved **AUC-ROC 0.9605, Accuracy 93%, F1-score 0.877** via 5-fold cross-validation with interpretable attention maps highlighting clinically relevant brain regions.

Beginner-Friendly Programming Chatbot | MERN Stack, Gemini API, Generative AI, Full-Stack | [GitHub](#)

- Built an **AI-powered chatbot** with **Google Gemini API** for beginner programmers; deployed full-stack using **MongoDB, Express.js, React.js, Node.js** with real-time conversational UI.
- Implemented **conversation context management** for multi-turn interactions and optimized **asynchronous API handling** for low-latency, scalable responses.

Waste Segregator Using OpenCV & YOLOv5 | Python, OpenCV, YOLOv5, Computer Vision | [GitHub](#)

- Developed a **real-time waste classification system** using **YOLOv5** achieving **90% detection accuracy**; processed live video with OpenCV for edge device deployment in smart bins.
- Applied **data augmentation and preprocessing** to improve model robustness across lighting/background variations; integrated **bounding-box visualization** and confidence scoring for interpretability.

TECHNICAL SKILLS

Languages: C, Python, JavaScript, SQL

Front-End: HTML, CSS, React.js, jQuery, Bootstrap

Back-End: Node.js, Express.js, Flask, Django, Firebase

Databases: MySQL, MongoDB

AI/ML: Machine Learning, Deep Learning, Computer Vision, NLP, Generative AI

Frameworks & Libraries: TensorFlow, Keras, PyTorch, OpenCV, Scikit-learn, YOLOv5/v8, Transformers

Developer Tools: Git, GitHub, Postman, Linux

ACHIEVEMENTS

- Ingenium 5.0 Hackathon – 1st Place:** Won national-level hackathon (theme: **Urban Traffic System**), designing an AI-powered intelligent traffic management solution.
- Technical Treasure Hunt – 1st Place:** Secured 1st in a multi-round challenge with **5 technical rounds** (image decoding via **OpenCV**) and **4 non-technical clue-solving rounds**.

CLUB ACTIVITIES

- BIS Club Coordinator** (Bureau of Indian Standards): Organizing awareness sessions on Indian standards, quality benchmarks, and standardization practices among students.
- CAMHI Club Coordinator** (Computer Aided Machine Human Interaction): Leading initiatives and events focused on HCI research, AI-driven interfaces, and human-computer interaction projects.